

Societal Problem: Housing in the Netherlands Week 2

September 2026

1 Scope of our Societal Problem

We want to focus mainly on the availability of housing rather than the affordability, which is why we modeled the renters without their income since this not only makes it easier but also addresses the more important issue of finding an apartment/house in the first place.

2 entities

As our entities we chose renters, landlords/contractor, contract, address, house and city.

2.1 relationships

3 attributes in each entity

people: PeopleID (int) (PK), first name (VARCHAR), last name (VARCHAR), age (int), address (FK)

renter: PeopleID (int) (FK, PK),

landlord/contractor: PeopleID (int) (FK, PK), houseID (that he manages) (FK)

contract: contractID (PK) (int), amount (int), start-date (DATETIME), end-date (DATETIME)

adress: Postalcode (int), cityID (FK), Street (VARCHAR), number (smallINT)

city: cityID (PK) (int), name (VARCHAR)

house: adressID (FK) (PK) (int), size (m^2) (int), contractID (FK) (int)

Entity	Relationship	Entity	Cardinality	Description
People	lives at	Address	Many-to-One	Many people can have the same address; each person has one address.
City	contains	Address	One-to-Many	One city can have many addresses; each address belongs to one city.
Address	identifies	House	One-to-One	Each house has one address, and each address represents one house in this model.
House	has	Contract	Many-to-One	A house has a contract; the contract can be associated with the house.
People	is a	Renter	One-to-One	Renter.People_PeopleID identifies a person.
People	is associated with	Landlord/Contractor	One-to-Many	One person can be associated with multiple houses.
House	is managed by	Landlord/Contractor	One-to-Many	One house can have multiple landlords/contractors.
People	is associated with	House	Many-to-Many	Implemented through Landlord/contractor , which acts as the junction table.

Table 1: Relationships between entities

4 ERD

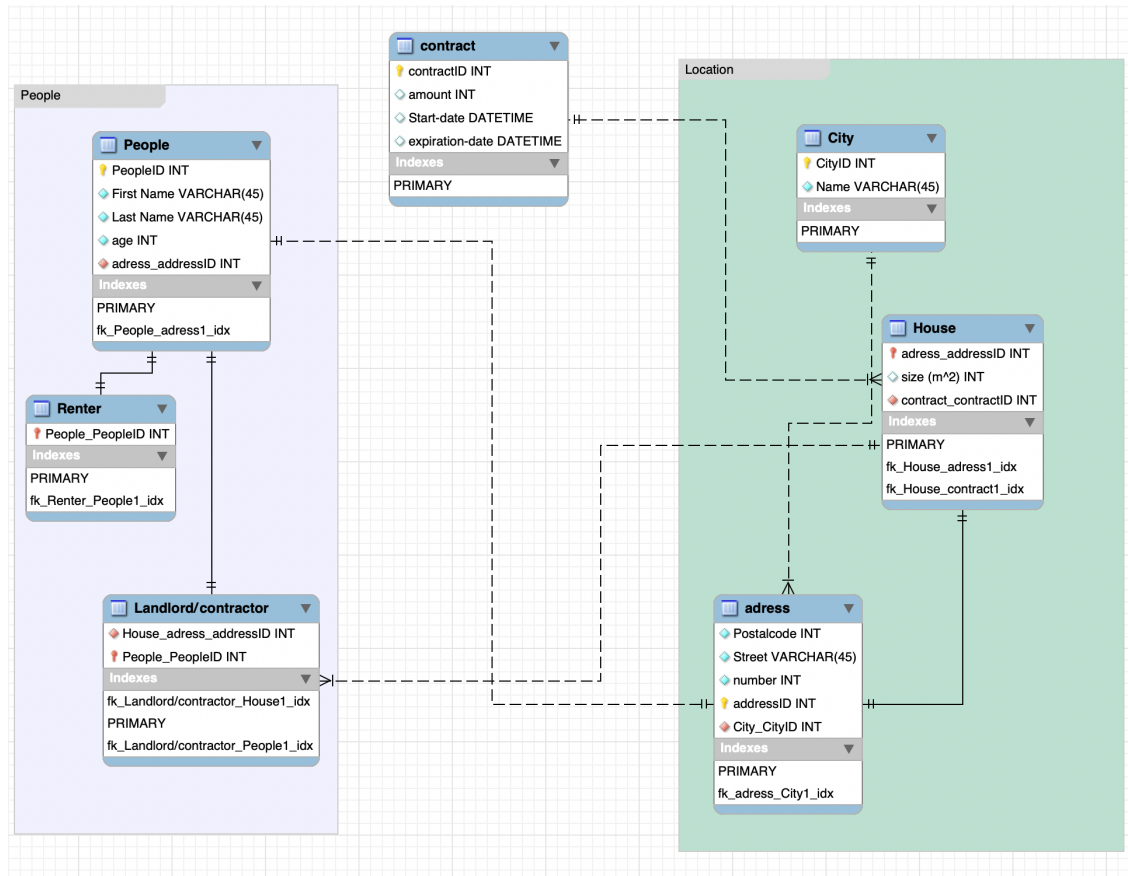


Figure 1: made in MySQL workbench

5 Normal Form (NF)

5.1 1NF

“Every cell must contain exactly one single, indivisible value (no lists or comma-separated groups), and there must be no repeating groups.”

As listed in the attributes chapter, there exists no list/comma-separated groups nor any repeating ones. (for example: address 1, address 2 would be a violation)

5.2 2NF

To be in 2NF, a database must be in 1NF. It should also have no partial dependencies, which means the non-key attribute cannot depend on only part of composite primary key. The *Landlord/contractor* table has composite primary key consisting of *PeopleID* and *addressID*. However, it has no other attributes. Therefore, there cannot be a partial dependency on only one part of

the key. The other tables have single-column primary keys such as *PeopleID*, *CityID*, *addressID*, and *contractID*. Therefore, partial dependencies are not possible in these tables.

5.3 3NF

To be in 3NF, a database must first be in 2NF. It should also have no transitive dependencies, which means a non-key attribute should not depend on another non-key attribute. In the *address* table, *addressID* determines the *Postcode*, *Street*, *number*, and *CityID*. The name of the city is not stored in the *address* table. Instead, it is stored in the separate *City* table, where *CityID* gives the city name.

This avoids a transitive dependency such as:

$$addressID \rightarrow CityID \rightarrow CityName$$

6 Comparison to a wrong ERD

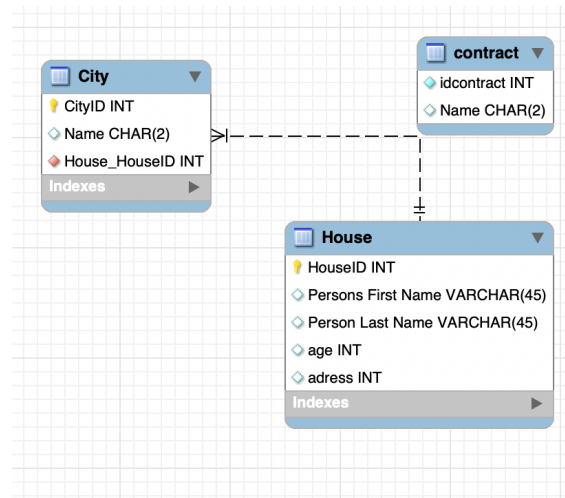


Figure 2: wrong ERD

Proposed Fixes:

-City: Name is in CHAR(2) meaning only 2 letter are allowed and expected which is not realistic. fix: change CHAR to VARCHAR(10-15) to allow different lengths of city names. House as a FK doesn't make any sense since a house doesn't have many cities, it has only one. fix: turn around the relation between City and house so each city has many houses.

-contract: no PK. fix: set idcontract to PK, name attribute makes no sense for contact. fix: remove name attribute

-house: violates 1NF since adress is a value containing a list of number, postalcode, street, city. Fix: detach adress into its own entity and add it in house as a FK. violation of 3NF since a person can own multiple houses so if the person ages we have to update all the houses he/she owns. Fix: detach Person and link it to house as a FK or even to address as a FK, if you are picky you could even separate between Renters and Landlords and create a superkey person.